Stylized Facts of the Implied Volatility Surface of Bitcoin Options

An Empirical Study of Maturity Effects, Skew, Risk Reversals and Butterfly Dynamics

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Abstract

This report studies stylized facts of the Bitcoin option implied volatility surface using two years of BTC option data. The project is empirical: implied volatilities are first introduced through the Black-Scholes inversion problem and then studied as market observables depending on strike, maturity and time. The first part recalls how implied volatility is computed, why the Black-Scholes flat-volatility assumption is contradicted by BTC option data, and how smile-aware delta hedging motivates the study of the volatility surface. The second part uses the SPX at-the-money skew as a benchmark and reviews power-law and multi-scale descriptions of the ATM skew term structure. The third part applies these ideas to BTC options and analyses risk reversals and butterflies across constant maturities and spot regimes. The main empirical finding is that the BTC smile is strongly maturity- and regime-dependent: short maturities display a pronounced downside skew on average, while longer maturities are closer to symmetric and can even become mildly call-skewed. Risk reversals are highly sensitive to BTC spot regimes, whereas butterflies mainly describe short-end smile curvature exploding for short maturities.

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1 Introduction

The Black-Scholes model gives a simple and internally consistent framework for pricing European options. Its main limitation, for the purpose of this project, is that the volatility parameter is assumed to be constant. If this assumption were true, all option prices on the same underlying would imply the same volatility, independently of strike and maturity. Empirically, this is not the case: implied volatility varies across strikes and maturities, forming a curved volatility surface.

This discrepancy is especially important for Bitcoin options. BTC is a highly volatile underlying asset, and its option market reflects both crash protection and speculative demand for upside convexity. Therefore, the implied volatility surface is not only curved, but also strongly time-varying. In particular, the sign and magnitude of the skew can change with the BTC spot regime. This makes BTC options an interesting laboratory for studying whether notions familiar from equity and foreign exchange options, such as downside skew, risk reversals and butterflies, behave differently in crypto markets.

The dataset used in the empirical part of the report contains SVI-implied volatility data for BTC options on Deribit, with observations from 1 January 2023 to 27 February 2025. In the current extraction, the file contains 188,277 observations, 8,903 timestamps and 788 calendar days. For each timestamp and maturity, it provides the ATM implied volatility, the implied volatilities of 10%, 20%, 30% and 40% out-of-the-money calls and puts, the underlying price, the BTC index price, the exchange and the time to expiration.

The first part introduces implied volatility, option price properties and the BTC volatility smile, with a link to the smile-adjusted delta hedging framework of Alexander and Imeraj [4]. The second part studies the SPX ATM skew as a reference case, using power-law and multi-scale parametric models inspired by the equity-index ATM-skew literature [5, 6]. The third part studies BTC risk reversals and butterflies, focusing on maturity dependence, time variation and spot-regime dependence. The interpretation is also compared with the stochastic-skew viewpoint for the foreign exchange market of Carr and Wu [3], where skew is understood as a time-varying state variable rather than a fixed deterministic curve.

The aim is to document and interpret empirical regularities of the BTC implied volatility surface: how the smile behaves across maturity, whether the BTC skew resembles the equity-index skew, how risk reversals react to spot regimes, and whether butterfly curvature has a simpler maturity structure.

2 Implied Volatility, BTC Option Smiles and the Volatility Surface

2.1 Black-Scholes option pricing and implied volatility

Let S_t denote the spot price of the underlying asset at time t, K the strike, T the maturity, r the constant risk-free rate, and σ the volatility parameter. We write

$$\tau = T - t$$

for the time to maturity. In the Black-Scholes model, the call option price is given by

$$C_t^{\mathrm{BS}} = S_t N(d_1) - K e^{-r\tau} N(d_2), \tag{1}$$

where

$$d_2 = \frac{\log(S_t/K) + \left(r - \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}, \qquad d_1 = d_2 + \sigma\sqrt{\tau}. \tag{2}$$

Equivalently,

$$d_1 = \frac{\log(S_t/K) + \left(r + \frac{1}{2}\sigma^2\right)\tau}{\sigma\sqrt{\tau}}. \tag{3}$$

Here N denotes the cumulative distribution function of a standard normal random variable:

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-u^2/2}\, \mathrm{d}u. \tag{4}$$

We also denote by N', or equivalently by φ, the standard normal density:

$$N'(x) = \varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}. \tag{5}$$

For reference, the corresponding Black-Scholes put price is

$$P_t^{\mathrm{BS}} = K e^{-r\tau} N(-d_2) - S_t N(-d_1). \tag{6}$$

For fixed S_t, K, T, and r, the implied volatility is defined as the value σ_{imp} satisfying

$$C_t^{\mathrm{mkt}} = C_t^{\mathrm{BS}}(\sigma_{\mathrm{imp}}). \tag{7}$$

The key mathematical point is that, under the usual no-arbitrage bounds (i.e the call price must lie between its intrinsic value and the underlying price $(S_t - Ke^{-r\tau})^+ \le C_t^{\mathrm{mkt}} \le S_t$.), this equation has a unique solution.

Indeed, the Black-Scholes call price is continuous and strictly increasing as a function of σ. Its derivative with respect to volatility is the Black-Scholes vega:

$$\frac{\partial C_t^{\mathrm{BS}}}{\partial \sigma} = S_t N'(d_1)\sqrt{\tau} > 0. \tag{8}$$

Moreover,

$$\lim_{\sigma \to 0^+} C_t^{\mathrm{BS}}(\sigma) = (S_t - Ke^{-r\tau})^+, \qquad \lim_{\sigma \to +\infty} C_t^{\mathrm{BS}}(\sigma) = S_t. \tag{9}$$

Therefore, the map $\sigma \longmapsto C_t^{\mathrm{BS}}(\sigma)$ is a bijection from $\mathbb{R}_+$ onto $\left((S_t - Ke^{-r\tau})^+, S_t\right)$. Thus, for any market call price satisfying the no-arbitrage constraint, there exists a unique implied volatility $\sigma_{\mathrm{imp}} > 0$.

In the numerical implementation, for simplicity, we take $S_0 = 1$ and assume $r = 0$. The Black-Scholes call price then reduces to

$$C_t^{\mathrm{BS}} = S_t N(d_1) - K N(d_2), \tag{10}$$

with

$$d_2 = \frac{\log(S_t/K) - \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \qquad d_1 = d_2 + \sigma\sqrt{\tau}. \tag{11}$$

In the pure Black-Scholes model, the volatility parameter σ is constant and does not depend on strike or maturity. Hence, if market prices were exactly generated by the Black-Scholes model, the inversion procedure would recover the same value for every pair (K, T):

$$\sigma_{\mathrm{imp}}(T, K) = \sigma. \tag{12}$$

Therefore, the Black-Scholes model predicts a flat implied volatility surface. The empirical observation of a volatility smile is precisely the failure of this constant-volatility assumption.

2.2 Basic properties of option prices

Before computing implied volatilities, it is useful to recall the main qualitative properties of European option prices. These properties are standard, but they are important because implied volatility is obtained by inverting option prices. Therefore, any numerical or empirical study of implied volatility first relies on understanding how option prices behave as functions of strike, maturity, spot price and volatility.

A European call option has payoff $(S_T - K)^+$, whereas a European put option has payoff $(K - S_T)^+$. For a call option, the price is increasing in the underlying price S_t and decreasing in the strike K. This is intuitive: a larger spot price makes the option more likely to finish in the money, whereas a larger strike makes the right to buy the asset less valuable. In the Black-Scholes model, these properties can be seen directly from

$$\frac{\partial C^{\mathrm{BS}}}{\partial S_t} = N(d_1) > 0, \qquad \frac{\partial C^{\mathrm{BS}}}{\partial K} = -e^{-r\tau} N(d_2) < 0. \tag{13}$$

Conversely, a put option is decreasing in the underlying price and increasing in the strike:

$$\frac{\partial P^{\mathrm{BS}}}{\partial S_t} = -N(-d_1) < 0, \qquad \frac{\partial P^{\mathrm{BS}}}{\partial K} = e^{-r\tau} N(-d_2) > 0. \tag{14}$$

Both call and put prices are increasing in volatility. Indeed, their derivative with respect to σ is the same Black-Scholes vega which was shown to be positive in the previous section by equation (8).

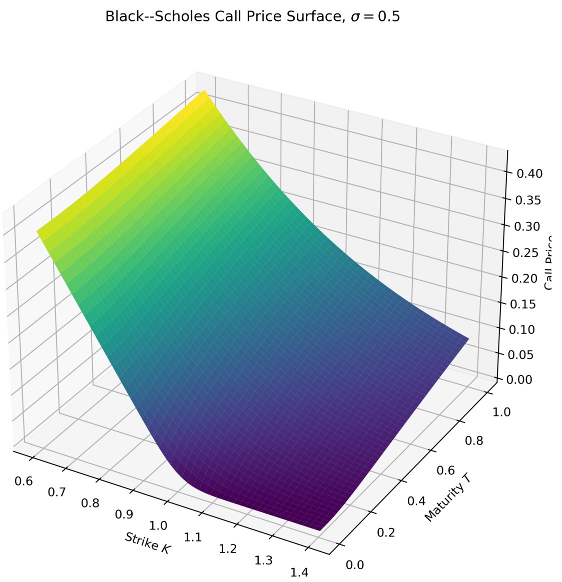
Another important property is convexity in strike. For a fixed maturity, option prices are convex functions of K. In the Black-Scholes model, for example,

$$\frac{\partial^2 C^{\mathrm{BS}}}{\partial K^2} = \frac{\partial^2 P^{\mathrm{BS}}}{\partial K^2} = \frac{e^{-r\tau} N'(d_2)}{K\sigma\sqrt{\tau}} > 0. \tag{15}$$

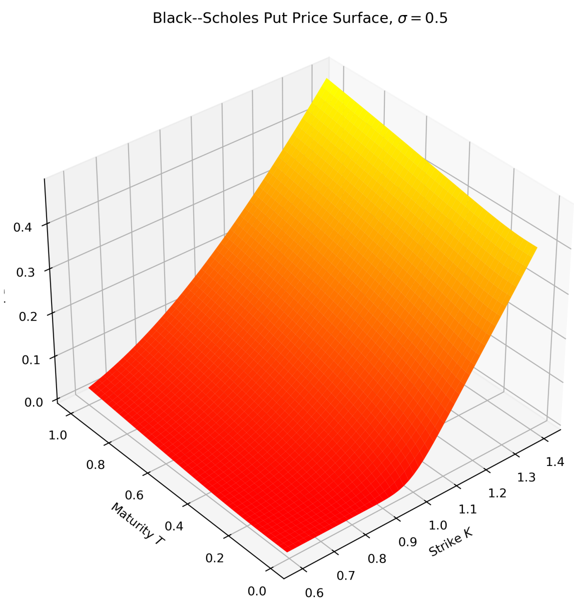
More generally, this convexity is linked to the risk-neutral distribution of the underlying at maturity: the curvature of option prices with respect to strike contains information about the probability distribution priced by the market. In particular, options with strikes far from the current underlying price correspond to tail events. A deep out-of-the-money call reflects the market-implied probability of a large upward move, while a deep out-of-the-money put reflects the probability of a large downward move. The smaller the probability assigned by the market to these tail scenarios, the lower the corresponding option price, although the price also depends on the size of the payoff conditional on reaching the tail.

These basic properties are first illustrated numerically by plotting Black-Scholes option price surfaces for a fixed volatility $\sigma = 0.5$. We take the same simple setting as in the numerical experiments, with $S_0 = 1$

and $r = 0$, and compute prices over a grid of maturities and normalized strikes. In this subsection, K should therefore be read as the strike-to-spot ratio, since $S_0 = 1$. Thus $K = 1$ corresponds to an at-the-money option, $K > 1$ corresponds to an out-of-the-money call, and $K < 1$ corresponds to an out-of-the-money put.



(a) Call option price surface.



(b) Put option price surface.

Figure 1: Black-Scholes call and put option price surfaces as functions of strike and maturity for a fixed volatility $\sigma = 0.5$.

The shapes of the two surfaces reflect the basic financial meaning of call and put options. For calls, the price is largest when $K < 1$, because the right to buy the asset below its current value is already valuable. As K increases above one, the call becomes out-of-the-money and its price decreases, since a larger upward move is required before maturity, for the call to have a positive payoff. Conversely, put prices increase with K, because a higher strike gives the holder the right to sell the asset at a more favourable price. In both cases, longer maturities generally increase the option value because they leave more time for the underlying to move into favorable regions. This time value is especially visible near the at-the-money region, where the final payoff is most uncertain.

The implied volatility surface is therefore an alternative representation of option prices, obtained through the Black-Scholes formula. In this framework, Black-Scholes is not used because its assumptions are believed to hold exactly. Rather, it is used as a market convention: implied volatility is the volatility input that makes the Black-Scholes formula match the observed option price. Mathematically, this interpretation follows from the positivity of vega, the volatility-derivative of option prices, as shown in equation (8). For fixed S_t, K, T, and r, volatility is the only free parameter in the Black-Scholes price. Hence, if a market option price is higher than the price produced by a benchmark volatility, the inversion can only match this price by increasing σ, which gives a higher implied volatility at that strike and maturity. This does not mean that the option is overvalued in an absolute sense. It means that, relative to the flat-volatility Black-Scholes benchmark, the market price embeds more uncertainty, or more weight on the relevant tail event. For example, a high implied volatility for an out-of-the-money put indicates that, under the market-implied risk-neutral distribution, downside tail events are priced more heavily than under the constant-volatility Black-Scholes lognormal distribution. Intuitively, this means that the market assigns more risk-neutral weight to scenarios in which BTC falls enough for the put option to finish in the money, so the put price is higher than what a flat-volatility Black-Scholes model would predict.

2.3 Numerical inversion methods

The previous subsection generated Black-Scholes call and put price surfaces with the known volatility $\sigma = 0.5$. We now use the call prices from this synthetic surface as input prices and try to recover the volatility that generated them. This reproduces, in a controlled setting, the same operation that will later be performed on real market option prices. The advantage is that the true volatility is known, so the accuracy and stability of the inversion methods can be compared directly.

For a fixed option with market price C^{mkt}, strike K, maturity T, spot price S_t, and interest rate r, we define

$$f(\sigma) = C^{\text{mkt}} - C^{\text{BS}}(S_t, K, T, r; \sigma). \tag{16}$$

In this numerical test, C^{mkt} is not yet a real market price: it is the synthetic call price previously computed with $\sigma = 0.5$. The implied volatility is then the unique root of f. Since the Black-Scholes call price is strictly increasing in σ, the function f is strictly decreasing. Therefore, once a bracket $[\sigma_{\text{min}}, \sigma_{\text{max}}]$ is found such that

$$f(\sigma_{\text{min}})f(\sigma_{\text{max}}) < 0,$$

the root is guaranteed to lie inside this interval.

A first naive implementation is the bisection method. Starting from a bracket $[\sigma_{\text{min}}, \sigma_{\text{max}}]$, one computes the midpoint

$$\sigma_{\text{mid}} = \frac{\sigma_{\text{min}} + \sigma_{\text{max}}}{2}. \tag{17}$$

If $f(\sigma_{\text{min}})f(\sigma_{\text{mid}}) < 0$, the root lies in the left subinterval and we replace $\sigma_{\text{max}} \leftarrow \sigma_{\text{mid}}$. Otherwise, the root lies in the right subinterval and we replace $\sigma_{\text{min}} \leftarrow \sigma_{\text{mid}}$. Repeating this procedure halves the length of the interval at each iteration. After n iterations, the bracket length is

$$\frac{\sigma_{\text{max}}^{(0)} - \sigma_{\text{min}}^{(0)}}{2^n}. \tag{18}$$

Bisection is therefore very robust and always converges if the initial bracket contains the root. Its drawback is that convergence is only linear, so it can require many iterations to reach high precision.

The second method is Newton–Raphson. It is obtained by taking a first-order Taylor expansion of f around the current iterate σ_n:

$$f(\sigma) \simeq f(\sigma_n) + f'(\sigma_n)(\sigma - \sigma_n). \tag{19}$$

To find an approximate zero, we set the right-hand side equal to zero at the next node σ_{n+1}:

$$0 = f(\sigma_n) + f'(\sigma_n)(\sigma_{n+1} - \sigma_n). \implies \sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)}. \tag{20}$$

which can be written as

$$\sigma_{n+1} = \sigma_n + \frac{C^{\text{mkt}} - C^{\text{BS}}(\sigma_n)}{\text{Vega}^{\text{BS}}(\sigma_n)}.$$

Newton's method is much faster than bisection when the initial guess is close enough to the solution, and it typically has quadratic convergence. However, it can be unstable when the vega is very small, for example for very short maturities or very deep in-the-money or out-of-the-money options. In that case, the Newton step can become too large, leave the positive volatility region, or diverge.

To combine stability and speed, a hybrid method can also be considered. The idea is to keep a valid bracket $[\sigma_{\text{min}}, \sigma_{\text{max}}]$, as in bisection, but to try a Newton step whenever possible. At iteration n, one first computes the Newton candidate σ_{Newt} given by σ_{n+1} in equation (20). If this candidate remains inside the current bracket ($\sigma_{\min} < \sigma_{\text{Newt}} < \sigma_{\max}$), then it is accepted. Otherwise, the method falls back to the bisection midpoint step. After the new point is chosen, the bracket is updated exactly as in the bisection method. This guarantees that the root is never lost, while allowing faster Newton steps when they are reliable.

Finally, Brent's method can be implemented as a derivative-free improvement of the same bracketing idea. It starts from a bracket containing the root, but instead of always taking the midpoint, it tries to predict the root using interpolation. The simplest interpolation step is the secant method. Given two points σ_1 and σ_2, we approximate f by the line passing through $(\sigma_1, f(\sigma_1))$ and $(\sigma_2, f(\sigma_2))$. The zero of this line is

$$\sigma_{\text{sec}} = \sigma_2 - f(\sigma_2)\frac{\sigma_2 - \sigma_1}{f(\sigma_2) - f(\sigma_1)}. \tag{21}$$

If three distinct points are available, Brent's method can use inverse quadratic interpolation. Instead of interpolating f as a function of σ, we interpolate the inverse relation $\sigma = \sigma(f)$ through the three points

$$(f(\sigma_1), \sigma_1), \qquad (f(\sigma_2), \sigma_2), \qquad (f(\sigma_3), \sigma_3).$$

Using the Lagrange interpolation formula for σ as a function of f, and evaluating it at $f = 0$, gives

$$\begin{aligned}\sigma_{\text{IQI}} = \sigma_1 &\frac{f(\sigma_2)f(\sigma_3)}{\big(f(\sigma_1) - f(\sigma_2)\big)\big(f(\sigma_1) - f(\sigma_3)\big)} \\ &+ \sigma_2 \frac{f(\sigma_1)f(\sigma_3)}{\big(f(\sigma_2) - f(\sigma_1)\big)\big(f(\sigma_2) - f(\sigma_3)\big)} \\ &+ \sigma_3 \frac{f(\sigma_1)f(\sigma_2)}{\big(f(\sigma_3) - f(\sigma_1)\big)\big(f(\sigma_3) - f(\sigma_2)\big)}.\end{aligned} \tag{22}$$

This is the candidate used when the three function values are distinct and the resulting point remains inside the current bracket. If this inverse quadratic step is not admissible, the method tries the secant step. If the secant step is also unsafe, it falls back to bisection. After each candidate is chosen, the bracket is updated so that the sign change is preserved. Hence Brent's method keeps the robustness of bisection while using faster interpolation steps whenever they are numerically reliable.

Finally, the recovered implied volatilities are compared to the true value $\sigma = 0.5$ used to generate the synthetic Black-Scholes call prices. The comparison is performed in two simple settings: first by varying the maturity T while keeping the strike K fixed, and then by varying the strike K while keeping the maturity T fixed. For each method, we report the maximum absolute error, the average absolute error, the average number of iterations, and the maximum number of iterations. This provides a controlled check of the numerical inversion procedure.

Table 1: Comparison of implied-volatility inversion methods on synthetic Black-Scholes call prices generated with true volatility $\sigma = 0.5$.

	Fixed K, varying T				Fixed T, varying K			
Method	Max. error	Avg. error	Avg. iter.	Max. iter.	Max. error	Avg. error	Avg. iter.	Max. iter.
Bisection	2.58×10^{-11}	2.58×10^{-11}	35.00	35	2.58×10^{-11}	2.58×10^{-11}	35.00	35
Newton–Raphson	5.02×10^{-10}	2.61×10^{-11}	3.84	4	2.61×10^{-10}	1.54×10^{-11}	4.94	7
Hybrid	9.31×10^{-10}	3.74×10^{-11}	2.80	3	4.48×10^{-10}	3.32×10^{-11}	3.58	5
Brent	3.75×10^{-10}	1.67×10^{-11}	3.92	5	9.94×10^{-10}	1.89×10^{-10}	9.58	25

The numerical comparison shows the expected trade-off between robustness and speed. Bisection is the safest method because it preserves a valid bracket at every step, but it requires the largest number of iterations. Newton–Raphson is much faster on average when it converges, although it can become unreliable when vega is small, especially for options far from the money. In some cases, Newton–Raphson did not return a valid implied volatility at all; these failures are not included in the performance table, which only reports errors and iterations for successful inversions. This is why its apparent speed must be interpreted with caution. The hybrid method combines the stability of bisection with the speed of Newton steps and gives the lowest average number of iterations in these tests. Brent's method also remains highly accurate and robust, although the fixed-maturity experiment shows that it may require more iterations for some strikes. Overall, this synthetic test is mainly used to justify the implied-volatility inversion framework before moving to the empirical analysis of the already-computed BTC implied volatility surface.

2.4 Empirical BTC implied volatility surfaces: moneyness, skew and first stylized facts

After the synthetic Black-Scholes tests, we now turn to the empirical BTC implied volatility data. In the previous numerical illustrations, we used the simplified setting $S_0 = 1$, so the strike K could be read directly as a normalized strike. We now return to the general framework, where the spot price is not normalized. The moneyness m and log-moneyness k are then defined by

$$m = \frac{K}{S_t}, \qquad k = \log(m) = \log\left(\frac{K}{S_t}\right). \tag{23}$$

The at-the-money region corresponds to $K \simeq S_t$, or equivalently $m \simeq 1$ and $k \simeq 0$. Out-of-the-money puts correspond to $K < S_t$, hence $k < 0$, while out-of-the-money calls correspond to $K > S_t$, hence $k > 0$.

For a fixed date t and maturity T, we denote the implied volatility smile by

$$k \longmapsto \sigma_{\mathrm{imp}}(t, k, T). \tag{24}$$

The log-moneyness skew is the slope of this smile with respect to k:

$$\partial_k \sigma_{\mathrm{imp}}(t, k, T). \tag{25}$$

In particular, the at-the-money skew is defined as

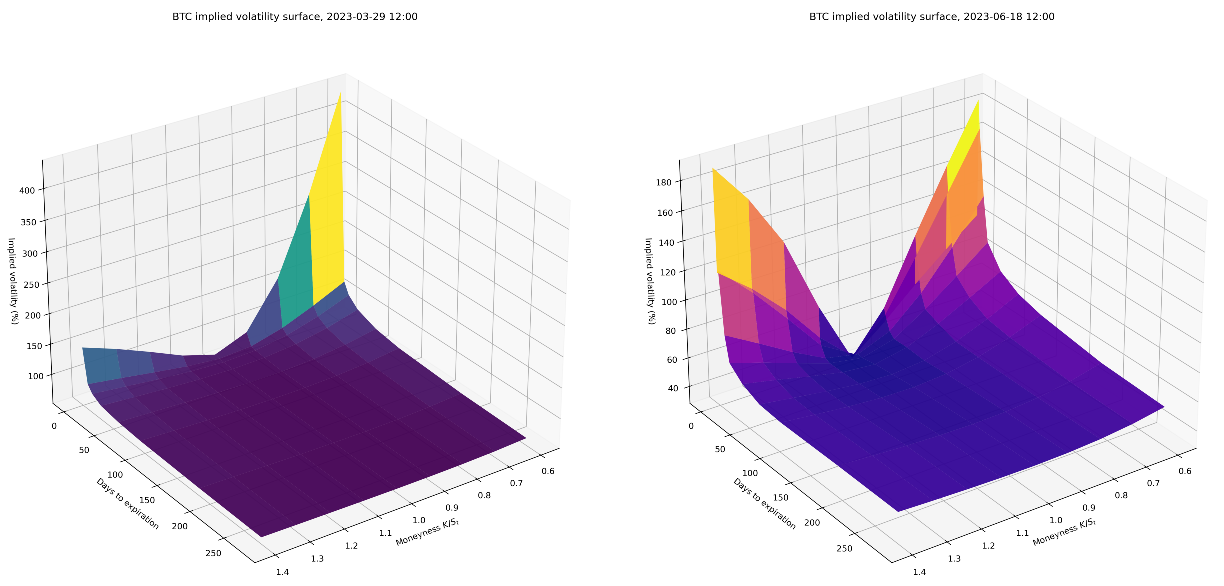
$$\mathrm{Skew}_{\mathrm{ATM}}(t, T) = \partial_k \sigma_{\mathrm{imp}}(t, k, T)\big|_{k=0}. \tag{26}$$

Since the dataset is observed on a discrete grid of moneyness levels, this derivative is approximated by finite differences around the ATM point. For example, using the 10% OTM call and put implied volatilities gives the proxy

$$\mathrm{Skew}_{10}(t, T) \approx \frac{\sigma_{\mathrm{imp}}\left(t, \log(1.10), T\right) - \sigma_{\mathrm{imp}}\left(t, \log(0.90), T\right)}{\log(1.10) - \log(0.90)}. \tag{27}$$

A negative value means that implied volatility decreases as log-moneyness increases around ATM. In financial terms, nearby OTM puts are then more expensive in implied-volatility terms than nearby OTM calls, which is usually associated with demand for downside protection. Conversely, a positive value means that the call side is more expensive, which can appear when the market prices upside convexity more aggressively.

A first visual inspection already shows that the BTC implied volatility surface is not stable over time. The dataset gives implied volatilities at standardized moneyness levels, from 40% OTM puts to 40% OTM calls, and across several maturities, depending on the date. The two surfaces below are taken from different dates and illustrate how the shape of the surface can change substantially over time.



(a) BTC IV surface on 29/03/2023.

(b) BTC IV surface on 18/06/2023.

Figure 2: Two examples of BTC implied volatility surfaces on different dates. The purpose is not yet to quantify the dynamics, but to show that the surface can change significantly across time.

Interactive animation of the BTC implied volatility surface over the full sample period:

`(Click here) BTC dynamic IV surface animation`

The animation makes visible the strong time variation of the surface over the two-year sample, and motivates the analysis of skew and smile dynamics in Section 4. We complement it here with a few static exploratory plots. The goal is not yet to give the final quantitative study of smile asymmetry and curvature, which will be done later, but rather to identify the main qualitative features of the BTC implied volatility surface.

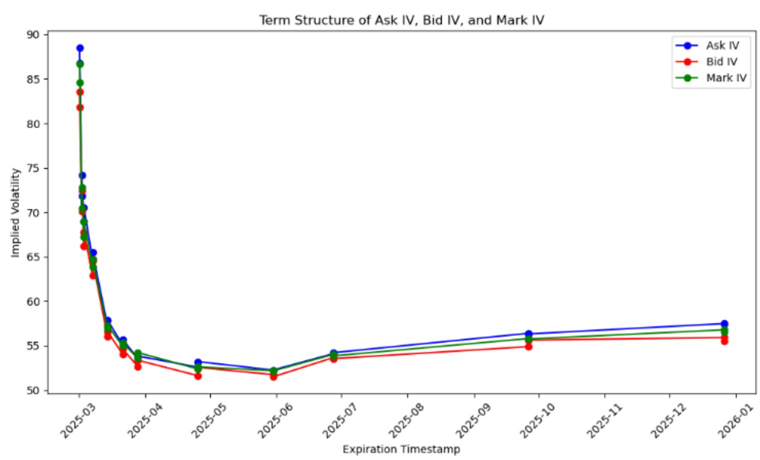


Figure 3: Term structure of BTC ask, bid and mark implied volatilities.

The first observation is that term structure of the IV level fluctuates significantly. Short maturities can have very high implied volatility because near-expiry options react strongly to immediate market moves. Mid-maturities may display lower implied volatility: they are not as exposed to very short-term shocks, but they are not long enough to embed a very wide range of possible BTC regimes. Longer maturities can rise again because they include more structural uncertainty and a broader set of possible future market scenarios. This reinforces the idea that implied volatility of the BTC must be studied as a surface, not only as a sequence of isolated smiles.

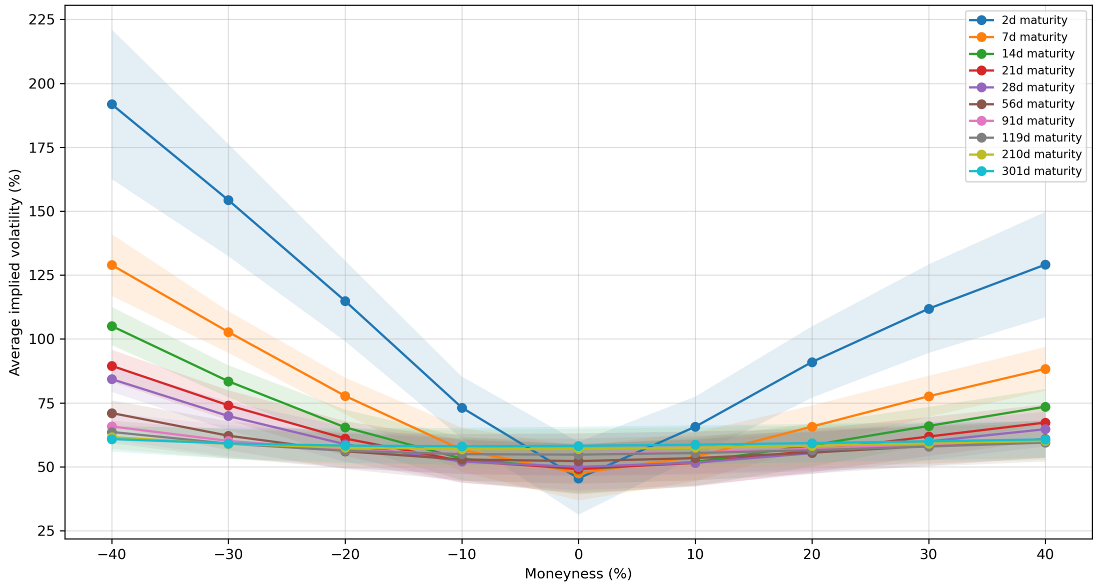
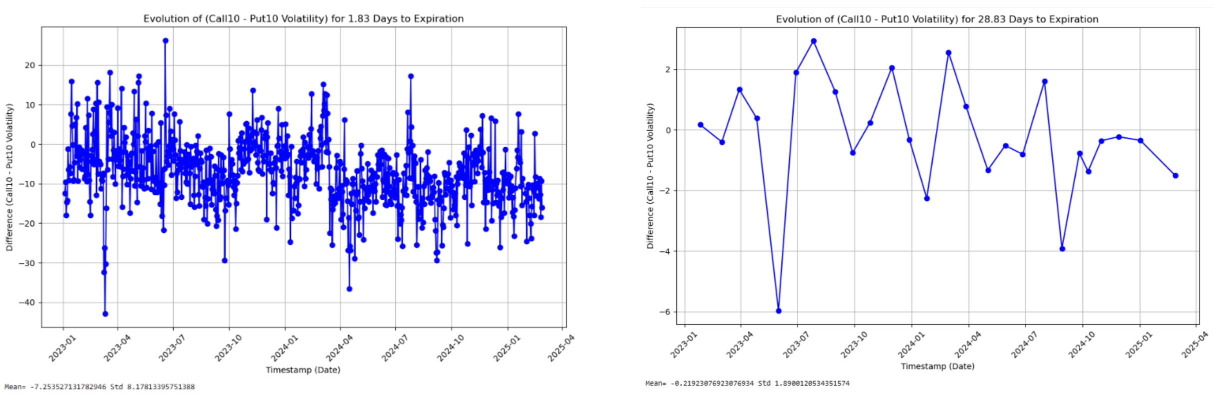


Figure 4: Average BTC implied volatility smiles across maturities with error bands.

Second, BTC options display a pronounced volatility smile. The smile is not constant across expirations: short-maturity smiles are generally steeper and more sensitive to the wings, while longer-maturity smiles are smoother. This already shows the full dependence of volatility on both moneyness and maturity.

To make this maturity effect more explicit, a first qualitative look at the difference between OTM call and OTM put implied volatilities is essential. At this stage, this is only used as a preliminary diagnostic for smile asymmetry. The detailed dynamic analysis will be postponed to the later section on smile asymmetry and curvature.

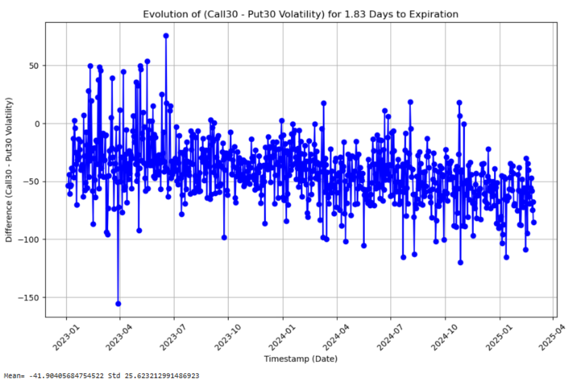


(a) 10% OTM call IV - 10% OTM put IV, short maturity. (b) 10% OTM call IV - 10% OTM put IV, longer maturity.

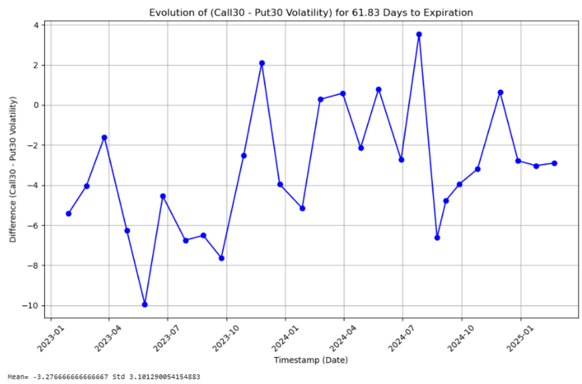
Figure 5: Preliminary comparison between 10% OTM call and put implied volatilities for two maturities.

These plots show that the difference between 10% OTM call IV and 10% OTM put IV is much more negative and more dispersed at short maturities. For example, the short-maturity case around 1.83 days to expiration has an average difference close to -7.25, with a standard deviation around 8.18. For a more intermediate maturity, the average difference is much closer to zero, around -0.22, with a smaller standard deviation around 1.89. For longer maturities, such as around 61.83 days, the average difference can even become slightly positive, around 0.53, with a standard deviation around 0.90. This indicates that the BTC skew is not only strong, but also very maturity-dependent, with OTM calls being sometimes more expensive than OTM puts for long maturities.

The same exploratory comparison can be pushed further into the wings of the smile by comparing 30% OTM calls and 30% OTM puts.



(a) 30% OTM call IV - 30% OTM put IV, short maturity.



(b) 30% OTM call IV - 30% OTM put IV, longer maturity.

Figure 6: Preliminary comparison between 30% OTM call and put implied volatilities for two maturities.

The 30% OTM comparison confirms the same qualitative effect, but with larger magnitudes. The far wings of the smile are more unstable than the near wings, especially for short-dated options. This means that the asymmetry of the BTC implied volatility surface is not only an at-the-money phenomenon: it becomes even more pronounced in the tails. In terms of smile parametrization, this corresponds qualitatively to stronger skewness and curvature at short maturities, followed by flatter and more symmetric/call-skewed surfaces at longer maturities.

Figure 7: BTC spot evolution with bullish rally periods highlighted, taken from TradingView.

The shape of the BTC smile is also connected to the market regime. In most short-maturity observations, OTM put implied volatilities are higher than OTM call implied volatilities, which is consistent with demand for downside protection. However, this effect is not permanent. During some strong bullish periods, the call side can become relatively more expensive, reflecting demand for upside convexity. This can already be seen qualitatively from the contrast between the two surfaces in Figure 2, and it is motivated by the rally periods shown in Figure 7. In particular, the periods where the call-minus-put differences become strongly positive tend to occur around bullish BTC phases, suggesting that upside convexity becomes more valuable when the spot market rallies. The precise link between spot regimes and smile asymmetry will be studied more systematically later.

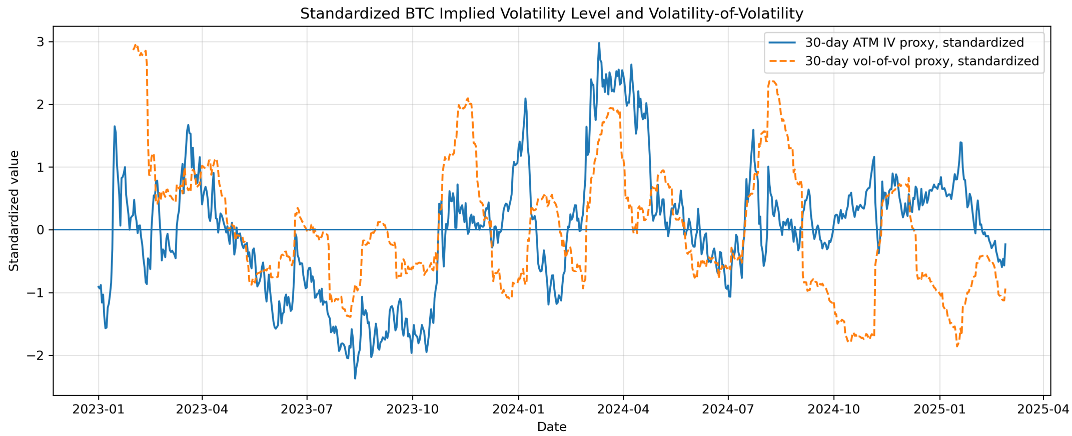


Figure 8: Standardized BTC 30-day ATM implied-volatility proxy and 30-day volatility-of-volatility proxy. Both series are standardized to compare their fluctuations on the same scale.

Finally, the exploratory analysis also points to strong fluctuations in the global level of BTC implied volatility. Instead of using a single quoted volatility, we construct a 30-day ATM implied-volatility proxy from the surface, which plays the role of a DVOL-like measure of the overall implied-volatility level. Figure 8 shows that this level varies substantially over time. The volatility-of-volatility proxy, computed from changes in this implied-volatility level, also displays clear spikes. This means that during turbulent periods the surface does not only shift upward; its level also becomes more unstable through time. This reinforces the main point of this first empirical view: BTC implied volatility is dynamic both in level and in shape, which is far from the constant-volatility Black-Scholes framework.

Overall, this first exploration shows that the BTC implied volatility surface is highly dynamic. It displays pronounced short-maturity skew, steep wings, maturity-dependent curvature, and substantial time variation. Compared with traditional markets, the BTC surface appears less stable: equity-index options usually exhibit a persistent negative skew, while FX smiles are often closer to symmetric. Bitcoin options combine features of both, but in a more regime-dependent way. Short maturities often display strong downside skew, as in equity-index markets, but they can also become near-symmetric or even call-skewed during some bullish episodes, reflecting demand for upside convexity. Longer maturities are generally flatter on average, but they also remain sensitive to changes in the market regime.

These observations motivate the need for pricing and hedging approaches that explicitly take the volatility smile into account. In particular, Alexander and Imeraj [4] study delta hedging of BTC options with a smile-adjusted volatility surface, showing that using a constant Black-Scholes volatility is not sufficient when the smile is strongly curved and time-varying.

Contribution of Part 1

The first part establishes the empirical and numerical framework of the study. We first introduced implied volatility through the Black-Scholes inversion problem and checked, on synthetic option prices, how classical numerical methods recover a known volatility. The BTC data showed that implied volatility depends strongly on moneyness, maturity, and time. It is a dynamic surface, with steep short-maturity wings, regime-dependent skew, unstable term structures, and fluctuations in the global volatility level. This motivates the rest of the study on the full surface.

3 ATM Skew Modelling: SPX as a Reference Case

3.1 Motivation: SPX as a reference market

Before studying BTC options, it is useful to begin with the SPX market as a reference case. SPX options form one of the most liquid and most studied option markets, and their implied volatility surface has a well-known and persistent structure: downside protection is usually priced more expensively than upside exposure. This makes the SPX a natural benchmark for understanding what a stable equity-index skew looks like before moving to a younger and more regime-dependent market such as BTC.

The goal of this part is therefore not to claim that BTC should behave like SPX. On the contrary, the SPX case provides a controlled reference against which BTC can later be compared. This comparison is useful because it helps separate general option-market effects, such as downside protection and maturity-dependent skew, from features that are more specific to the crypto market and to BTC in particular.

3.2 ATM skew and its term structure

We recall that the ATM skew is the derivative of implied volatility with respect to log-moneyness at the money as written in equation (26). As this quantity is typically negative for equity indices such as the SPX, the ATM skew will be studied in absolute value, using average market data from 2012 to 2023. The function of interest is therefore $S_{ATM}(T) = \partial_k \sigma_{iv}(T, k) \mid_{k=0}$, which is the term structure of the ATM skew. The data here included the exact derivative value, a finite difference approximation was therefore not needed.

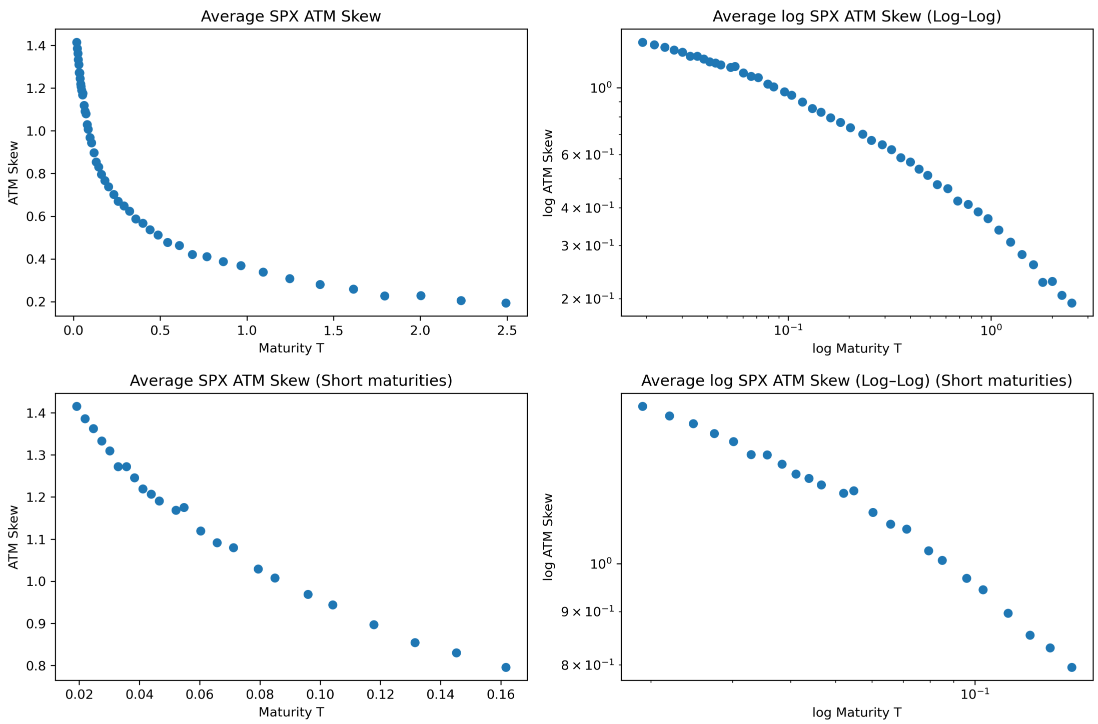


Figure 9: Average SPX ATM skew term structure in standard scale and log-log scale, with focus on short maturities.

The empirical shape is clear. The ATM skew is largest at short maturities and decreases as maturity increases. On a log-log plot, part of the curve appears approximately linear, suggesting a power-law-type decay:

$$|S_{ATM}(T)| \approx CT^{-\alpha}. \tag{28}$$

However, the log-log plot is not perfectly linear across the full maturity range. This suggests that a single power law is too rigid, and that different maturity regimes may be present:

$$|S_{ATM}(T)| \approx \begin{cases} C_1 T^{-\alpha_1}, & T \text{ short}, \\ C_2 T^{-\alpha_2}, & T \text{ long}. \end{cases} \tag{29}$$

This first observation justifies using SPX as a reference case before returning to BTC. The SPX market provides a benchmark where the negative skew is persistent and economically well understood, since it is usually linked to demand for downside protection and crash-risk premia. However, Figure 9 also shows that even in this mature and liquid market, the ATM skew term structure is not described perfectly by a single power law. The short end and the longer end of the curve appear to follow different behaviours, which motivates a two-regime or multi-scale description. This point is important for the comparison with BTC: if a relatively stable equity-index market already requires a non-trivial maturity-dependent model, then the more unstable and regime-dependent BTC surface should not be expected to follow a simple one-regime structure. Studying SPX first therefore helps to extract features that are more specific to crypto markets, such as stronger regime dependence and possible changes in the sign of the skew.

3.3 Parametric calibration of the ATM skew term structure

We now calibrate several parametric functions to the full SPX ATM skew term structure. Since the empirical SPX ATM skew is negative, the calibration is performed on its absolute value,

$$S(T) = |S_{\mathrm{ATM}}(T)|.$$

The goal is to compare simple one-scale models with more flexible specifications able to capture the curvature observed in Figure 9.

The first model is a single-scale exponential-inspired shape:

$$S_{\exp}(T) = \frac{c_1}{T}\left(1 - \frac{1 - e^{-\lambda_1 T}}{\lambda_1 T}\right), \qquad c_1, \lambda_1 > 0. \tag{30}$$

This form introduces one characteristic maturity scale through λ_1, but remains relatively rigid.

The second model is a fractional power law:

$$S_{\mathrm{frac}}(T) = cT^{\beta}, \qquad \beta > -0.5. \tag{31}$$

In log-log scale, this model becomes exactly linear.

The third model is a shifted power-law-type specification:

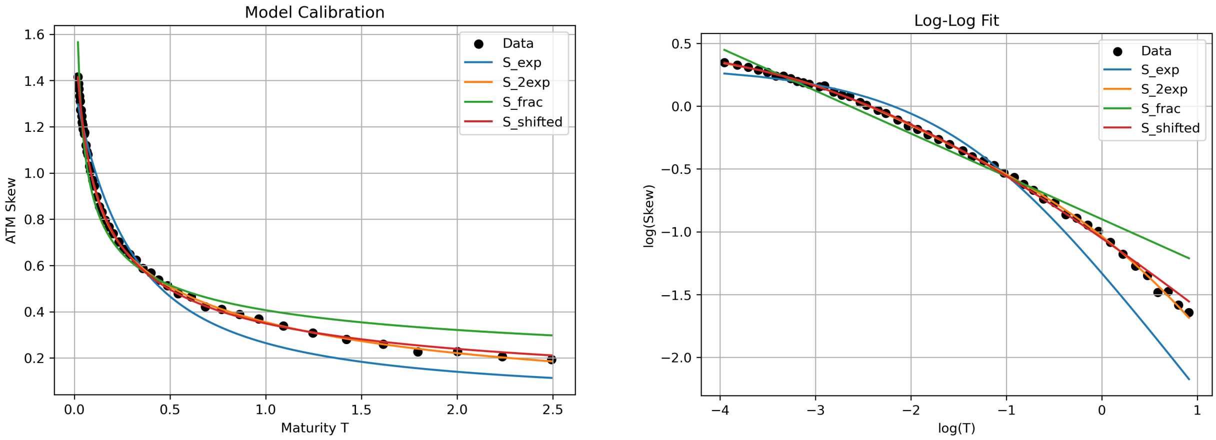
$$S_{\mathrm{shifted}}(T) = \frac{c}{T^2}\left(\frac{(\alpha + T)^{\beta+2}}{\beta+2} - \frac{\alpha^{\beta+2}}{\beta+2} - \alpha^{\beta+1}T\right), \qquad \beta > -1. \tag{32}$$

In the calibration, the shift is fixed at $\alpha = \frac{1}{52}$, so that the model keeps only two free parameters. This shift regularizes the short end and gives the model more flexibility than a pure power law.

Finally, we also calibrate a two-exponential model:

$$S_{2\exp}(T) = \frac{c_1}{T}\left(1 - \frac{1 - e^{-\lambda_1 T}}{\lambda_1 T}\right) + \frac{c_2}{T}\left(1 - \frac{1 - e^{-\lambda_2 T}}{\lambda_2 T}\right), \qquad c_i, \lambda_i > 0. \tag{33}$$

This model introduces two characteristic time scales. It is therefore closer in spirit to multi-factor volatility models, where a fast factor captures the short end and a slower factor controls longer maturities.



(a) Standard scale.

(b) log-log scale.

Figure 10: Calibration of the four parametric models to the full SPX ATM skew term structure.

Table 2: Goodness-of-fit statistics for the full SPX ATM skew calibration. Lower SSE, RMSE, MAE, AIC and BIC indicate better fit, while higher R^2 and adjusted R^2 indicate better explanatory power.

Model	n_{params}	SSE	MSE	RMSE	MAE	R^2	Adj. R^2	AIC	BIC
$S_{2\,\text{exp}}$	4	2.993×10^{-3}	6.236×10^{-5}	7.896×10^{-3}	5.580×10^{-3}	0.999574	0.999534	-456.769	-449.284
S_{shifted}	2	5.781×10^{-3}	1.204×10^{-4}	1.098×10^{-2}	8.517×10^{-3}	0.999176	0.999140	-429.167	-425.425
S_{frac}	2	1.723×10^{-1}	3.590×10^{-3}	5.991×10^{-2}	5.063×10^{-2}	0.975449	0.974358	-266.226	-262.483
S_{exp}	2	2.063×10^{-1}	4.299×10^{-3}	6.557×10^{-2}	5.703×10^{-2}	0.970599	0.969292	-257.572	-253.830

The calibration confirms that a pure fractional power law captures the broad decreasing trend of the SPX ATM skew, but is too rigid on the full maturity range. The single-scale exponential model is also too restrictive: it introduces one characteristic scale, but does not reproduce the observed maturity dependence accurately enough.

Among the two-parameter models, S_{shifted} gives the best fit. The fixed shift $\alpha = 1/52$ acts as a regularization of the short end and allows the model to preserve a power-law-type behavior while adapting better to the curvature of the data. The best global fit is obtained with $S_{2\,\text{exp}}$. It has two additional parameters, but the improvement is large enough to be supported by both AIC and BIC. This suggests that the SPX ATM skew term structure is better described by a multi-scale mechanism than by a single power law.

3.4 Comparison with the literature

The calibration results are consistent with the empirical literature on the SPX ATM skew term structure. Guyon and El Amrani [5] study whether the equity-index ATM skew can really be described by a pure power law. Their conclusion is that a power law captures the main scaling behaviour away from the very short end, but becomes less accurate once short maturities are included. To correct this, they introduce regularized power-law specifications, such as the time-shifted power law

$$\Lambda_{\text{TSPL}}(t, u, y) = c\,(u - t + \delta)^{-\alpha} y_u, \tag{34}$$

and the capped power law

$$\Lambda_{\text{CPL}}(t, u, y) = c\ \max(u - t, \delta)^{-\alpha} y_u. \tag{35}$$

Both preserve the global power-law behaviour while smoothing or capping the short end. They also compare these models with the two-factor Bergomi-type form

$$\Lambda_{2\text{fB}}(t,u,y) = \left(c_1 e^{-k_1(u-t)} + c_2 e^{-k_2(u-t)}\right) y_u, \tag{36}$$

which combines a fast and a slow time scale. This is directly related to the calibration above: the fractional model S_{frac} plays the role of the pure power law, while S_{shifted} and especially $S_{2\,\text{exp}}$ introduce extra flexibility at the short end and across maturities.

The work of Delemotte, De Marco and Ségonne [6] is particularly relevant here because it refines the power-law interpretation of the SPX ATM skew. Using a longer SPX sample and including weekly options, they obtain a richer description of the short-maturity region. Their main empirical finding is that the average ATM skew is not well described by one straight line on a log-log plot. Instead, they identify two regimes: a short-maturity regime with a milder slope, approximately $\alpha_1 \simeq -0.3$, and a longer-maturity regime closer to the classical exponent $\alpha_2 \simeq -0.5$, with a transition around $T_1 \simeq 2$ months.

To describe this crossover, they introduce a smooth two-power-law model of the form

$$S_T^{2\text{PL}} = S_{T_1}\left(\frac{T}{T_1}\right)^{\alpha_1}\left(\frac{\left(\frac{T}{T_1}\right)^{\kappa}+1}{2}\right)^{\frac{\alpha_2-\alpha_1}{\kappa}}. \tag{37}$$

Here T_1 is the transition maturity, S_{T_1} is the skew level at that transition, α_1 and α_2 are the short- and long-maturity slopes, and κ controls how smooth the transition is. This model is important conceptually because it keeps a power-law interpretation, but allows the exponent to change across maturities.

This comparison clarifies the role of the models calibrated above. The shifted model improves the fit by regularizing the short end, while the two-exponential model improves it by introducing two characteristic time scales. In this sense, $S_{2\,\text{exp}}$ is not exactly a two-power-law model, but it captures the same empirical idea: the SPX ATM skew term structure is multi-scale.

Contribution of Part 2

The second part provides a solid reference before returning to BTC. The SPX case shows that even in a mature and liquid equity-index market, the ATM skew term structure requires regularized or multi-scale models to describe its maturity dependence. This is important because BTC is expected to be even less stable: its skew can change sign, its wings are more volatile, and its shape reacts strongly to spot regimes. The next section therefore moves from this SPX reference case to the dynamic analysis of BTC smile asymmetry and curvature.

4 Risk Reversal and Butterfly Dynamics of BTC Options

4.1 Smile asymmetry and curvature indicators

To quantify the dynamics of the BTC implied volatility smile, we use two standard smile diagnostics: risk reversals and butterflies. They are computed directly from the implied volatility surface at fixed maturity T and date t, using symmetric out-of-the-money call and put volatilities.

For $q \in \{10, 20, 30, 40\}$, the q% risk reversal is defined by

$$\mathrm{RR}_q(T,t) = \sigma^{\mathrm{call}}_{q\%\mathrm{OTM}}(T,t) - \sigma^{\mathrm{put}}_{q\%\mathrm{OTM}}(T,t). \tag{38}$$

It measures the directional asymmetry of the smile. A negative risk reversal means that OTM puts have higher implied volatility than OTM calls, which is usually interpreted as stronger demand for downside protection. Conversely, a positive risk reversal means that OTM calls are richer in implied-volatility terms, reflecting stronger pricing of upside convexity.

The q% butterfly is defined by

$$\mathrm{BF}_q(T,t) = \frac{1}{2}\left(\sigma^{\mathrm{call}}_{q\%\mathrm{OTM}}(T,t) + \sigma^{\mathrm{put}}_{q\%\mathrm{OTM}}(T,t)\right) - \sigma_{\mathrm{ATM}}(T,t). \tag{39}$$

Unlike the risk reversal, the butterfly is not directional. It compares the average implied volatility of the two wings with the ATM implied volatility, and therefore measures the curvature of the smile. A high butterfly indicates that tail options are expensive relative to ATM options, independently of whether the expensive wing is on the put side or the call side.

Finally, the risk reversal can be related to a finite-difference approximation of the ATM skew. Since q% OTM puts and calls correspond approximately to log-moneyness levels $\log(1-q/100)$ and $\log(1+q/100)$, we define

$$\mathrm{SKEW}_q(T,t) = \frac{\mathrm{RR}_q(T,t)}{\log(1+q/100) - \log(1-q/100)}. \tag{40}$$

This quantity approximates the slope of the implied volatility smile around the ATM point. It provides the link between the ATM-skew analysis of the previous section and the risk-reversal dynamics studied here.

4.2 Construction of a daily constant-maturity panel

The raw BTC option dataset contains observations at many non-identical maturities at each timestamp. Since risk reversals, butterflies and skew proxies must be compared through time at the same maturity, the first step is to construct a constant-maturity panel.

Before choosing the target maturities, we inspect the empirical distribution of days to expiration. The purpose is not simply to keep the maturities with the largest absolute number of observations, since these would be concentrated almost entirely at the very short end. Instead, the target grid is chosen to cover the whole maturity range while staying close to regions where the dataset has sufficient local density. This reduces interpolation error while preserving information on short, medium and long expiries.

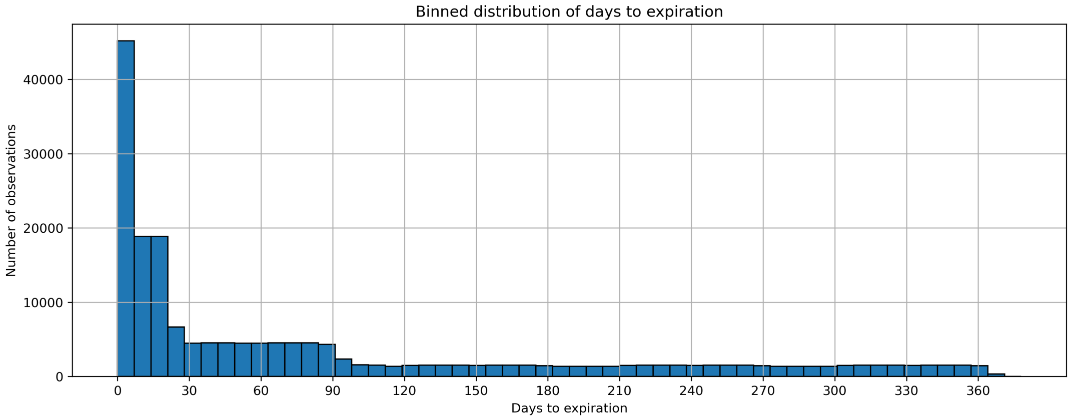


Figure 11: Binned distribution of days to expiration in the BTC option dataset. The target maturities are chosen near locally well-populated maturity regions, while still covering the full maturity range.

The selected target maturities are

$$T \in \{5, 7, 12, 20, 30, 42, 56, 74, 90, 120, 150, 180, 240, 270, 330, 360\} \text{ days.} \tag{41}$$

For each timestamp t, the available maturities are first sorted by days to expiration. Then, for each target maturity T^*, every quantity of interest is linearly interpolated across maturity. If $T_j \le T^* \le T_{j+1}$, then for any metric X, such as ATM IV, RR_q, BF_q, or SKEW_q, we set

$$X(t, T^*) = \frac{T_{j+1} - T^*}{T_{j+1} - T_j} X(t, T_j) + \frac{T^* - T_j}{T_{j+1} - T_j} X(t, T_{j+1}). \tag{42}$$

No extrapolation is used: if the target maturity lies outside the range of available maturities at a given timestamp, the value is left missing.

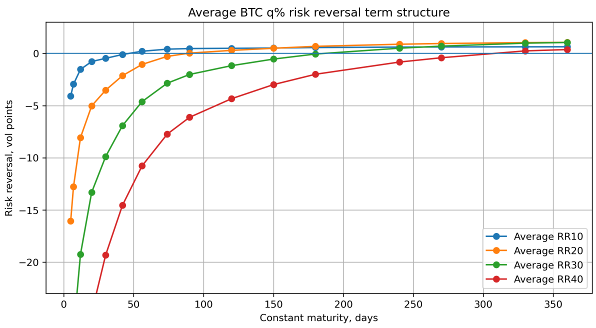
Finally, the interpolated intraday panel is aggregated to daily frequency by averaging observations with the same calendar date and target maturity:

$$\overline{X}(d, T^*) = \frac{1}{N_{d,T^*}} \sum_{t \in d} X(t, T^*). \tag{43}$$

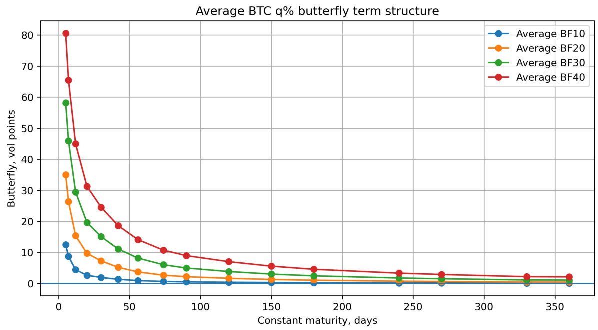
This produces a daily constant-maturity panel of ATM implied volatility, risk reversals, butterflies and finite-difference skew proxies. This panel is the common empirical basis for the rest of the analysis.

4.3 Average BTC smile asymmetry and curvature across maturities

We now study the average term structure of BTC risk reversals and butterflies across the constant-maturity panel. For each target maturity, the quantities are averaged over all dates in the sample. This gives a first static summary of the BTC smile dynamics before looking at time variation and spot-regime effects.



(a) Average q% risk reversal term structure.



(b) Average q% butterfly term structure.

Figure 12: Average BTC risk reversal and butterfly term structures for $q \in \{10, 20, 30, 40\}$.

The average risk reversal term structure is strongly maturity-dependent. At short maturities, all risk reversals are negative on average, meaning that OTM puts are richer than OTM calls in implied-volatility terms. This is the equity-like part of the BTC smile: near-expiry downside protection is priced more strongly than upside convexity. However, the negative skew weakens quickly with maturity. For longer maturities, the average risk reversals move closer to zero and can even become mildly positive, indicating that the long end of the BTC surface is much less persistently put-skewed than the SPX reference case.

The comparison across moneyness levels is also informative. The 10% risk reversal describes the near-ATM asymmetry, while the 30% and 40% risk reversals measure the far-wing asymmetry. The further we move into the wings, the more negative the short-maturity risk reversal becomes. This shows that the BTC downside skew is not only an ATM phenomenon: it is amplified in the tails, especially close to expiry. At longer maturities, the different q% curves become much closer, which suggests that the long-end smile is flatter and less directionally asymmetric.

The butterfly term structure behaves differently. For all moneyness levels, the average butterfly is positive, meaning that the two wings are more expensive than the ATM point on average. This confirms that BTC options display a genuine smile curvature, not only a directional skew. The butterfly is largest at short maturities and decays rapidly as maturity increases. The effect is again stronger for larger q: BF_{30} and BF_{40} are much larger than BF_{10} at the short end, because far-wing options embed more tail-risk premium.

Overall, the two plots separate two different effects. Risk reversals show that the direction of the BTC smile changes with maturity: short maturities are strongly put-skewed on average, while longer maturities are closer to symmetric or mildly call-skewed. Butterflies show that smile curvature is always positive but concentrated at the short end. This distinction will be important in the rest of the section: risk reversals are mainly used to study directional smile asymmetry and regime dependence, while butterflies measure the intensity of tail-risk pricing independently of direction. However, this first view is only based on unconditional averages; the picture becomes more informative once the same quantities are studied conditional on specific BTC spot regimes later in the section.

4.4 Focus on modeling of BTC finite-difference ATM skew

We now return to the finite-difference ATM-skew proxy introduced above in equation (40). This quantity plays, for BTC, the same role as the ATM skew studied in the SPX reference case. However, its empirical behaviour is fundamentally different. In the SPX market, the ATM skew remains negative across maturities, so it is natural to study the decay of its absolute value and to compare one-regime and two-regime power laws. For BTC, the average SKEW_{10} is strongly negative at short maturities but crosses zero around

$$T^* \simeq 46.5 \text{ days},$$

before becoming mildly positive at longer maturities. Therefore, separating the curve into two negative power-law regimes, as in the SPX analysis, is not appropriate. Moreover, there are relatively few maturity points before the zero crossing, so fitting a separate short-end power law would be fragile.

For this reason, we model the full signed term structure directly. The objective is to capture the transition from short-maturity downside skew to a flatter, approximately symmetric or mildly call-skewed long end. Just like for the SPX market, we fit transition functions of the form $T \longmapsto S(T)$, where $S(T)$ denotes the average SKEW_{10} at maturity T. Among the tested specifications, we retain the best-performing models that remain reasonably parsimonious.

The first retained model is a stretched exponential transition,

$$S(T) = L - A\exp\left[-\left(\frac{T}{\tau}\right)^p\right]. \tag{44}$$

This model allows a rapid short-end adjustment followed by convergence to a long-maturity level L. The second is a rational transition,

$$S(T) = L - \frac{A}{(T+\theta)^b}. \tag{45}$$

This specification is flexible at the short end while allowing slow long-maturity flattening. The third is a saturating logarithmic transition,

$$S(T) = L - \frac{A}{1 + b\log\left(1+\frac{T}{\theta}\right)}. \tag{46}$$

This logarithmic form is useful because it can describe a fast initial adjustment while still converging slowly at longer maturities. More flexible generalized versions were also tested, but are not retained here because the improvement in fit was minimal compared to the cost of an additional parameter.

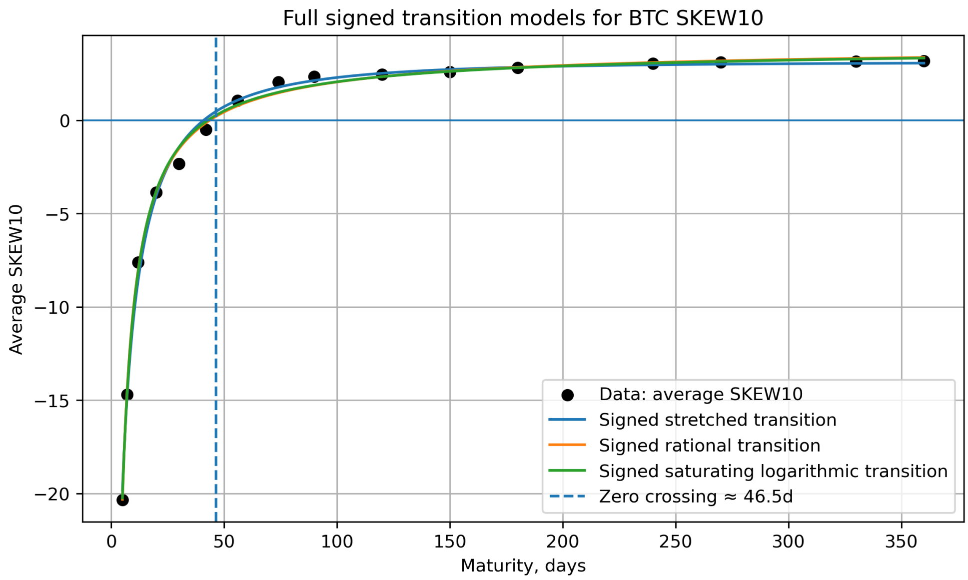


Figure 13: Signed transition models fitted to the average BTC SKEW_{10} term structure.

Table 3: Goodness-of-fit statistics for the retained models fitted to the BTC SKEW_{10} term structure.

Signed Model	n_{params}	SSE	MSE	RMSE	MAE	R^2	Adj. R^2	AIC	BIC
Rational transition	4	1.656811	0.103551	0.321793	0.228545	0.997771	0.996961	-28.283	-25.193
Saturating logarithmic transition	4	1.777348	0.111084	0.333293	0.231922	0.997609	0.996740	-27.159	-24.069
Stretched transition	4	2.533214	0.158326	0.397902	0.280015	0.996593	0.995353	-21.490	-18.399

The best fit is obtained with the signed rational transition. It has the lowest RMSE and MAE, the highest adjusted R^2, and the best information criteria among the retained models. The saturating logarithmic transition performs almost as well and gives a similar qualitative shape, while the stretched transition captures the broad transition but is less accurate around the curvature of the curve.

Financially, the rational transition gives a natural description of the BTC skew term structure. It captures a very negative short end, corresponding to expensive short-dated downside protection, followed by a rapid move toward symmetry and then a slower flattening at longer maturities. This confirms the main difference with SPX: BTC skew is not simply a persistent negative skew whose magnitude decays with maturity, but a signed maturity transition from put-skewed short maturities to a more symmetric or mildly call-skewed long end. From a behavioral point of view, this can be interpreted as follows: the short end is dominated by immediate crash-risk protection, while the medium-to-long end can incorporate demand for upside convexity, consistent with speculative interest in large BTC rallies; at even longer horizons, the smile becomes closer to symmetric because both positive and negative long-run scenarios remain plausible and are priced into the two wings.

4.5 Smile regimes and BTC spot regimes

The unconditional averages studied above show that the BTC smile is strongly maturity-dependent, but they do not fully describe how often the smile is actually put-skewed, near symmetric, or call-skewed through time. We therefore first classify the 10% risk reversal into three simple smile regimes:

$$\begin{cases} \text{put-skewed}, & \mathrm{RR}_{10} < -1, \\ \text{near symmetric}, & |\mathrm{RR}_{10}| \le 1, \\ \text{call-skewed}, & \mathrm{RR}_{10} > 1. \end{cases} \tag{47}$$

The threshold of one volatility point is only used to avoid treating very small numerical deviations from zero as meaningful directional skew.

Table 4: Smile-regime frequencies for selected maturities. Entries are percentages of days.

Maturity (days)	Call-skewed	Near symmetric	Put-skewed
5	16.9	8.6	74.5
7	20.2	12.2	67.6
30	25.0	37.7	37.3
90	30.8	60.7	8.5
180	25.3	71.7	3.0
360	1.3	98.4	0.4

The frequencies confirm the maturity pattern observed in the average term structure. At very short maturities, the smile is put-skewed most of the time: for example, the 5-day maturity is put-skewed about 74.5% of the time. As maturity increases, the smile becomes much closer to symmetric. Around 90 days, the near-symmetric regime becomes dominant, and at 360 days the smile is almost always classified as near symmetric. This confirms that the persistent negative skew is mainly a short-end phenomenon.

To quantify the relation between smile behaviour and BTC market conditions, rather than relying only on the qualitative observations made in section 2.4, we classify the BTC spot regime using recent 30-day log-returns $r_{30}(t)$ and 30-day return persistence $p_{30}(t)$, defined as the fraction of positive daily returns over the previous 30 days:

$$r_{30}(t) = \log S_t - \log S_{t-30} \qquad p_{30}(t) = \frac{1}{30}\sum_{j=1}^{30} \mathbf{1}_{\{r_{1d}(t-j+1)>0\}} \tag{48}$$

The spot regimes are then defined by

$$\text{spot regime}(t) = \begin{cases} \text{consistent rally}, & r_{30}(t) > 0.15 \text{ and } p_{30}(t) \geq 0.65, \\ \text{jumpy rally}, & r_{30}(t) > 0.15 \text{ and } p_{30}(t) < 0.65, \\ \text{selloff}, & r_{30}(t) < -0.15, \\ \text{neutral}, & \text{otherwise.} \end{cases} \tag{49}$$

These thresholds were selected after a grid search over a range of return and persistence cutoffs; the qualitative regime patterns described below remain stable across nearby parameter choices, so the results are not sensitive to the specific values chosen. Still, this classification is purely descriptive. Its purpose is to separate persistent rallies, less regular upward moves, selloffs and neutral periods.

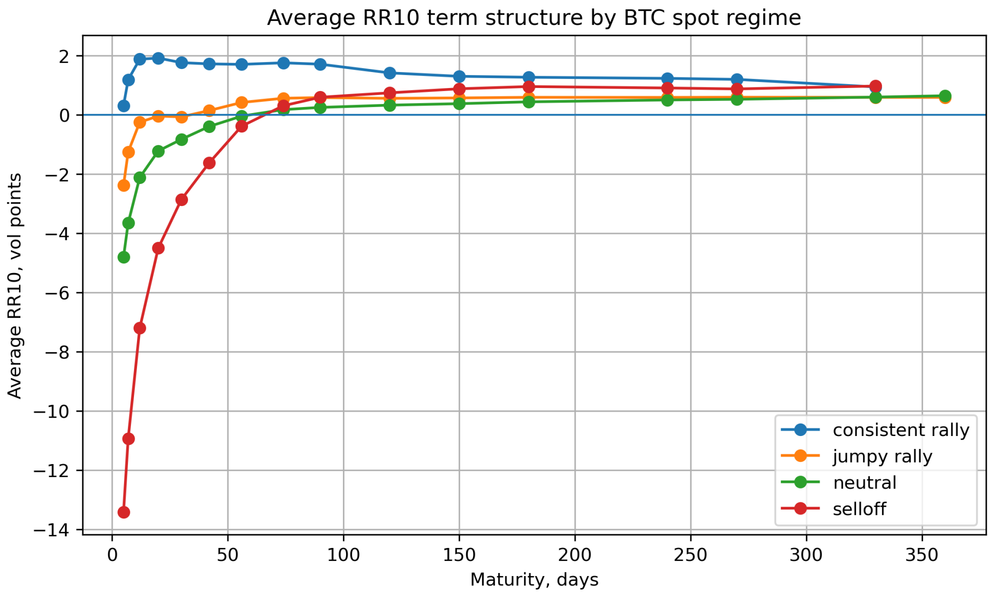
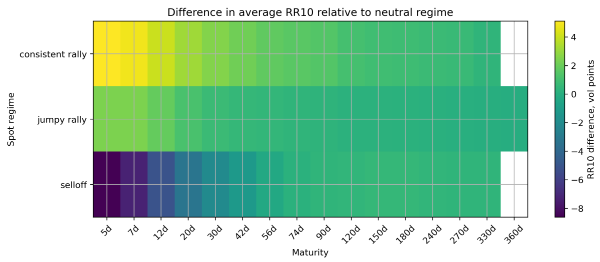


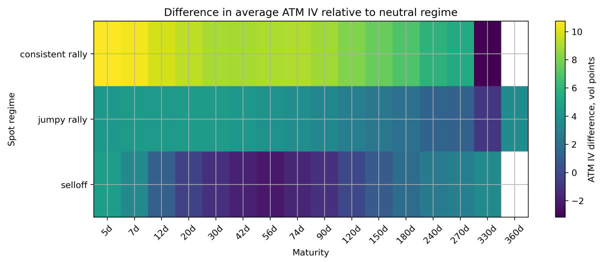
Figure 14: Average BTC RR_{10} term structure by spot regime. Persistent rallies shift the risk reversal upward, while selloffs make the short end more negative.

The regime decomposition gives a clearer economic interpretation than the unconditional average. During consistent rallies, the RR_{10} term structure is already positive at the shortest available maturities. This means that, in these periods, even short-dated BTC options become call-skewed on average: upside calls are richer than downside puts in implied-volatility terms. This is the clearest evidence that BTC differs from the standard equity-index case. During selloffs, the opposite effect appears: the short-end risk reversal becomes sharply negative, indicating a strong repricing of downside protection. Neutral periods remain negative at the short end but move toward symmetry with maturity, while jumpy rallies lie between neutral periods and consistent rallies.

The zero-crossing maturity also depends on the spot regime: it occurs earlier in rally regimes and is pushed further out as the regime becomes more bearish, reflecting the stronger persistence of short-end downside skew in selloffs. However, all regimes gradually converge toward a near-neutral RR_{10} at longer maturities, suggesting that the long-term BTC smile becomes less dependent on the current spot regime and returns closer to a balanced outlook.



(a) RR_{10} relative to neutral.



(b) ATM IV relative to neutral.

Figure 15: Differences in average RR_{10} and ATM implied volatility relative to neutral periods.

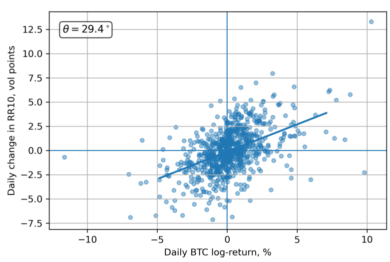
The heatmaps confirm the regime effect described above, but in relative terms with respect to neutral periods. Consistent rallies shift RR_{10} upward across most maturities, with the strongest difference at the short end, while selloffs shift it downward and reinforce the short-maturity put skew. The effect fades as maturity increases, which is consistent with the convergence of all regimes toward a more neutral long-maturity risk reversal. The ATM IV heatmap adds a second dimension: consistent rallies are also associated with higher ATM implied volatility relative to neutral periods. Thus BTC rallies in this sample are not simply quiet low-volatility rallies; they can coincide with both a higher ATM volatility level and stronger demand for upside convexity.

4.6 BTC spot trends as drivers of RR_{10}

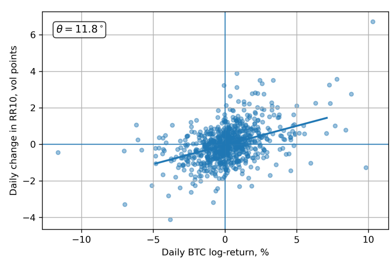
The regime classification above shows that the shape of the BTC smile depends on the recent state of the spot market. We now look more directly at what drives this regime dependence. The aim is to distinguish two effects: the impact of short-term BTC returns, and the impact of more persistent trends over several days or weeks, and which horizon is the main driver.

A first diagnostic is the contemporaneous relation between daily BTC returns and daily changes in RR_{10}. For each maturity T where $r_{1d}(t)$ is the daily BTC log-return, we consider

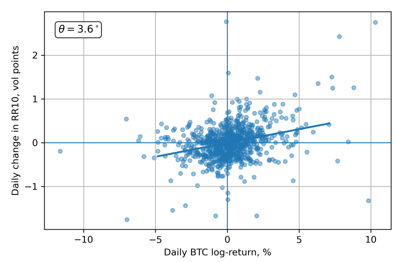
$$\Delta \mathrm{RR}_{10}(T,t) = a_T + b_T r_{1d}(t) + \varepsilon_{T,t}, \tag{50}$$



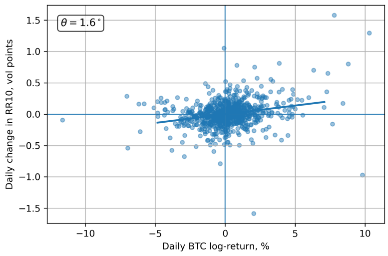
(a) $T = 7$ days.



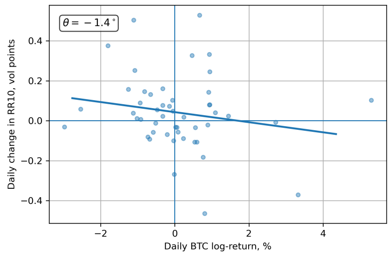
(b) $T = 30$ days.



(c) $T = 90$ days.



(d) $T = 180$ days.



(e) $T = 360$ days.

Figure 16: Daily BTC returns and daily changes in RR_{10} for selected maturities. The angle θ measures the tilt of the fitted regression line relative to the x-axis. As maturity increases, the fitted slope becomes flatter, indicating a weaker contemporaneous relation between spot returns and changes in RR_{10}.

The scatter plots show that the contemporaneous relation is strongest at short and medium maturities. For 7-day and 30-day maturities, positive BTC returns are often associated with positive changes in RR_{10}: the call wing becomes richer relative to the put wing. Negative returns tend to move the smile in the opposite direction. As maturity increases, the cloud of points progressively loses this upward orientation and becomes more symmetric around the origin, with an overall 'clockwise rotation' of this cloud. This means that daily spot moves have a much weaker direct effect on long-maturity risk reversals.

The second diagnostic studies instead whether the overall level of RR_{10} is related to recent BTC trends and return persistence, and in particular which time horizon appears most relevant. A natural first specification would be to regress RR_{10} on several return and persistence horizons at the same time:

$$\mathrm{RR}_{10}(T,t) = a_T + \sum_{h\in\{5,7,20,30,90\}} \left[b_{T,h} r_h(t) + c_{T,h} p_h(t) \right] + \varepsilon_{T,t}. \tag{51}$$

Here $r_h(t)$ is the BTC log-return over the previous h days and $p_h(t)$ is the fraction of positive daily returns over the same window. However, because these horizons are still overlapping, estimating all of them in the same regression can make the comparison between horizons less transparent, due to the covariance between the different indicators. We therefore use a simpler descriptive diagnostic: for each maturity and each horizon, we estimate separate univariate regressions,

$$\mathrm{RR}_{10}(T,t) = a_{T,h} + b_{T,h} r_h(t) + \varepsilon_{T,h,t}, \tag{52}$$

and separately

$$\mathrm{RR}_{10}(T,t) = \alpha_{T,h} + c_{T,h} p_h(t) + \eta_{T,h,t}. \tag{53}$$

This isolates each return and persistence driver one at a time. The goal is not to identify causality, but to compare which horizons have the strongest descriptive association with the level of RR_{10}, using the t-statistic and R^2 as summary measures.

Table 5: Dominant return and persistence horizons in univariate regressions for RR_{10}.

Maturity	Return driver	Coef.	t-stat	R^2	Persistence driver	Coef.	t-stat	R^2
5	r_{20d}	0.2683	18.81	0.3370	p_{30d}	30.9979	19.01	0.3419
7	r_{20d}	0.2496	19.84	0.3612	p_{30d}	29.0364	20.28	0.3714
20	r_{20d}	0.1508	20.03	0.3656	p_{30d}	17.1841	19.82	0.3609
30	r_{20d}	0.1117	19.26	0.3477	p_{30d}	12.6975	18.99	0.3414
90	r_{20d}	0.0453	14.89	0.2417	p_{30d}	5.1338	14.64	0.2355

The univariate regressions give a very stable result across maturities. Among return variables, the strongest association is always obtained with the 20-day BTC return. The positive coefficients mean that when BTC has risen over the previous 20 days, RR_{10} tends to be higher, so the smile shifts toward the call wing. Among persistence variables, the dominant horizon is always p_{30d}, the fraction of positive BTC days over the previous 30 days. This shows that RR_{10} is related not only to the size of the recent BTC move, but also to how consistently positive the recent path has been.

The strength of the relation decreases with maturity. For the return driver, the coefficient decreases from 0.2683 at 5 days to 0.0453 at 90 days, and the R^2 decreases from about 0.34–0.37 at short and medium maturities to about 0.24 at 90 days. The same pattern appears for persistence. This confirms that BTC spot trends mainly affect short- and medium-maturity smile asymmetry. Financially, the result is consistent with the regime analysis: when BTC has been rising over several weeks, especially with persistent positive days, upside convexity becomes more valuable and RR_{10} shifts upward. During weaker or negative trends, the market moves back toward downside protection.

4.7 Butterfly behaviour across regimes and maturity

The analysis above shows that RR_{10} is highly directional and strongly related to spot regimes. The butterfly has a different interpretation. It measures the average richness of the two wings relative to ATM, and is therefore less directly linked to the sign of the BTC move.

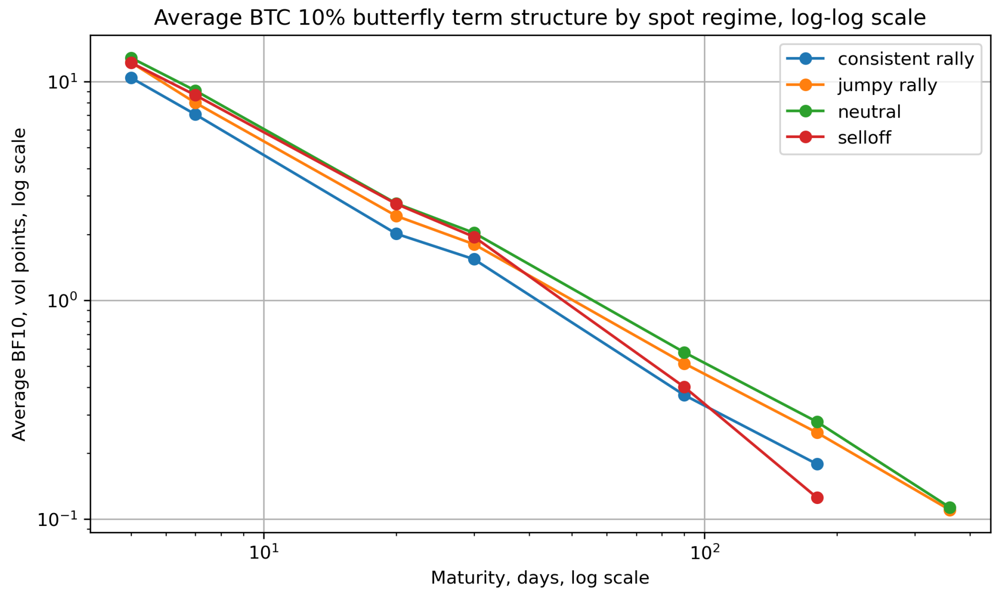


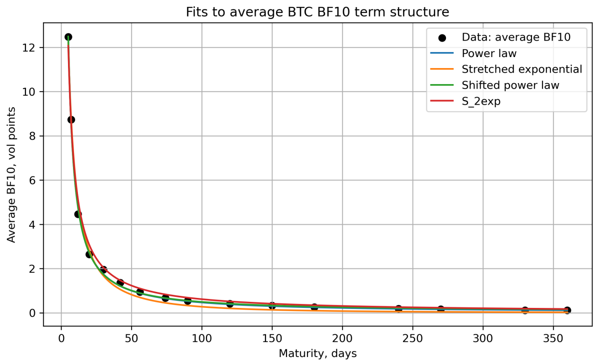
Figure 17: Average BTC BF_{10} term structure by spot regime in log-log scale. The butterfly remains positive and decreases linearly with maturity (in log-log scale) in all regimes.

The main information in Figure 17 is not the level of the butterfly itself, which was already discussed in section 4.3, but the much weaker regime separation compared with RR_{10}. The curves remain close across spot regimes and preserve almost the same decreasing shape in log-log scale. This suggests that BF_{10} is mainly driven by maturity and tail-risk richness, rather than by the direction of the current BTC spot regime. In contrast, RR_{10} reacts much more clearly to rallies and selloffs because it measures the relative price of the call wing versus the put wing.

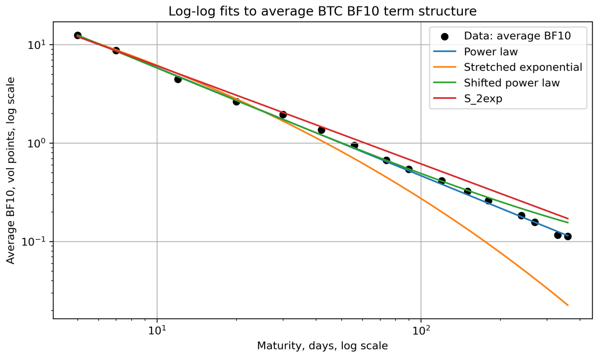
The log-log shape is particularly informative. The curves are close to straight lines from short to medium maturities, suggesting a power-law-type decay of the average butterfly term structure. To test this, we fit several positive maturity-decay models to the unconditional average BF_{10} term structure, including respectively a simple power law and a shifted power law:

$$\mathrm{BF}_{10}(T) = AT^{-\beta}, \qquad \mathrm{BF}_{10}(T) = L + \frac{A}{(T+\theta)^{\beta}}. \tag{54}$$

Exponential-type alternatives were also used, namely the stretched exponential and the two-exponential model introduced respectively in equation (44) and equation (33). The fitted models are reported in Figure 19:



(a) Standard scale.



(b) log-log scale.

Figure 18: Average BTC BF_{10} term structure and power-law maturity-decay fits. The log-log representation shows that the decay is close to linear, especially at medium and long maturities.

Table 6: Average BF_{10} term-structure fit quality.

Model	Params	SSE	RMSE	MAE	R^2	Adj. R^2	AIC	BIC
Shifted power law	4	0.178389	0.105590	0.068846	0.999052	0.998708	-63.942	-60.852
Power law	2	0.191100	0.109287	0.066955	0.998985	0.998829	-66.841	-65.296
$S_{2\,\mathrm{exp}}$	4	0.876238	0.234019	0.169049	0.995345	0.993652	-38.475	-35.385
Stretched exponential	3	0.987175	0.248392	0.216395	0.994755	0.993444	-38.568	-36.250

The shifted power law gives the lowest raw error, but the improvement over the simple power law is very small. The simple power law uses only two parameters, still gives an excellent R^2 and adjusted R^2, and obtains the best AIC and BIC. It is therefore the preferred specification by parsimony. This is also visible in the log-log plot: the simple power law follows the medium- and long-maturity points particularly well, where the decay of BF_{10} is almost linear. With additional long-maturity observations, this behaviour would likely make the simple power law even more favourable.

The exponential-type alternatives are less convincing. Although $S_{2\,\mathrm{exp}}$ and the stretched exponential still give high R^2 values, they over-steepen the decay at the ends of the maturity range, especially in log-log scale. The stretched exponential in particular bends downward too strongly at long maturities, which is not consistent with the observed decay of the butterfly. For this reason, these models are useful as diagnostics but are not retained as the main description.

This contrasts with RR_{10}. The risk reversal changes sign and is strongly regime-dependent, so signed transition models are needed. The butterfly remains positive and decays smoothly with maturity, so a simple power-law decay already captures its average term structure very well. In other words, BTC smile asymmetry is complex and state-dependent, while BTC smile curvature has a simpler average maturity structure.

4.8 Interpretation through stochastic skew

The stochastic-skew perspective of Carr and Wu [3] provides a useful interpretation of the whole section, not only of the butterfly analysis above . The previous subsections showed that RR_{10} is strongly state-dependent, while BF_{10} has a more stable maturity-decay structure. This suggests that the direction of the smile should not be viewed as a fixed deterministic feature of the BTC option market. Instead, the skew behaves like a time-varying state variable, changing with market conditions and especially with the recent BTC spot regime.

Carr and Wu make a similar point in the FX option context: the average risk-neutral distribution can be close to symmetric, while the conditional distribution on a given date can become strongly asymmetric and can even change direction through time. This is close in spirit to what is observed here for BTC options. The average long-maturity RR_{10} is close to symmetric or mildly positive, but the regime analysis shows that short maturities can become heavily put-skewed during selloffs and call-skewed during persistent rallies. The positive contemporaneous relation between BTC returns and ΔRR_{10} reinforces this interpretation: rallies tend to move the risk reversal upward, while selloffs move it downward.

The BTC results should therefore not be interpreted as saying that RR_{10} mechanically predicts future rallies. The more careful conclusion is that BTC option markets price a time-varying conditional asymmetry. When upside convexity becomes more valuable, RR_{10} rises; when crash protection dominates, RR_{10} becomes negative. This is precisely why a stochastic-skew perspective is more appropriate than a purely static implied-volatility surface.

Contribution of Part 3

The third part gives the main empirical contribution of the study. Butterflies mainly measure short-end curvature and follow a simpler power-law decay. BTC risk reversals, by contrast, reveal a strong maturity and regime dependence: short-dated options are put-skewed on average, but persistent rallies can shift the smile toward call skew. Together, these results show that the BTC volatility surface is best understood as a dynamic object, not as a static smile.

5 Conclusion

5.1 Key Takeaways

This research studied stylized facts of the Bitcoin option implied volatility surface using two years of BTC options data. The starting point was the Black-Scholes definition of implied volatility: volatility is not taken as a structural constant, but as the value that makes the Black-Scholes option price match the observed market price.

The first part of the report showed that BTC implied volatility must be studied as a surface. Short maturities can exhibit steep wings and strong skew, while longer maturities are generally smoother. The surface also changes substantially across dates, as illustrated by the static examples and by the supplementary animation. The 30-day ATM implied-volatility proxy and the volatility-of-volatility proxy further showed that the BTC volatility level itself is unstable. These observations already suggested that BTC options cannot be described by a single volatility number.

The second part used the SPX market as a reference case. In the SPX market, ATM skew is persistently negative and is usually interpreted as the pricing of downside protection and crash-risk premia. However, the SPX term structure is already richer than a single power law: regularized or multi-scale models are needed to describe the full maturity range. This benchmark is useful because it separates general option-market effects from features that are more specific to BTC.

The main empirical contribution came from the analysis of BTC risk reversals and butterflies. Risk reversals measure the directional asymmetry of the smile, while butterflies measure its non-directional curvature. The average BTC risk reversal term structure is strongly maturity-dependent: short maturities are put-skewed on average, but this negative skew weakens quickly and the long end becomes close to symmetric or mildly call-skewed. The finite-difference skew proxy therefore changes sign, unlike the SPX ATM skew. For this reason, BTC skew is better described by signed transition models than by positive power-law decay models. Among the fitted specifications, the rational transition gave the most convincing description of the average SKEW_{10} term structure.

The regime analysis gives a financial interpretation of the BTC smile. In equity-index markets, negative skew is usually linked to crash protection, but BTC shows that this logic is not permanent. When BTC is in a persistent rally, investors are not only hedging downside risk; they also appear willing to pay for upside convexity. This pushes RR_{10} upward and can make short-dated calls richer than puts. During selloffs, the market returns to a more classical protection-driven structure: short-dated puts become expensive because investors price the risk of further sharp downside moves. The return-driver analysis supports this interpretation: RR_{10} is not only related to one-day spot moves, but also to recent BTC trend and persistence, namely around 20-day returns and 30-day return persistence. Financially, this suggests that the smile reacts not only to isolated jumps, but to the formation of a sustained market regime. The fact that these effects fade with maturity indicates that spot regimes mainly affect the near-term conditional distribution of BTC, while longer maturities price a wider range of possible future scenarios.

Butterflies give a different financial message. Since BF_{10} averages the two wings, it is less sensitive to whether the market is bullish or bearish. It mainly measures how expensive tail exposure is relative to ATM volatility. The fact that butterflies remain positive and decay smoothly with maturity suggests that BTC options consistently price large moves on both sides, especially near expiry. However, this tail-risk premium has a simpler maturity structure than the directional skew. This is why a simple power law can describe the average BF_{10} term structure well, whereas RR_{10} needs signed, regime-dependent models. In financial terms, the market's view on the direction of extreme BTC moves is unstable, but the richness of short-dated tail exposure is a more persistent feature of the BTC options surface.

Overall, the study shows that the BTC implied volatility surface is best understood as a dynamic state variable of the market. The results are also consistent with a stochastic-skew interpretation: the average smile may look close to symmetric at longer maturities, while the conditional smile on a given date can be strongly asymmetric and can change sign with the market regime.

5.2 Limitations and possible extensions

The main limitation of the study is that it is descriptive. The spot-regime classification is mechanical and the constant-maturity panel relies on interpolation across available maturities. Moreover, the dataset covers a specific 2-year market period starting in January 2023 and ending in February 2025.

Natural extensions would be to repeat the analysis on future BTC data and on other crypto underlyings, in order to distinguish stylized facts specific to Bitcoin from broader crypto-option features. Another direction would be to test stochastic volatility or stochastic-skew models that are consistent with the observed behaviour: short-end downside skew, regime-dependent risk reversals, and smoother butterfly decay. Finally, one could study the impact of specific crypto-market events, such as ETF announcements, exchange failures, regulatory news or major liquidity shocks, on the level, skew and curvature of the BTC implied volatility surface.

References

[1] F. Black and M. Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

[2] R. C. Merton. Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

[3] P. Carr and L. Wu. Stochastic skew in currency options. *Journal of Financial Economics*, 86(1):213–247, 2007.

[4] C. Alexander and A. Imeraj. Delta hedging bitcoin options with a smile. *Quantitative Finance*, 23(5):799–817, 2023. DOI: 10.1080/14697688.2023.2181205.

[5] J. Guyon and M. El Amrani. Does the term-structure of equity at-the-money skew really follow a power law? SSRN working paper, 2022.

[6] J. Delemotte, S. De Marco and F. Ségonne. Yet another analysis of the SP500 at-the-money skew: crossover of different power-law behaviours. SSRN working paper, 2023.